MIZORAM PUBLIC SERVICE COMMISSION

COMPETITIVE EXAMINATION FOR RECRUITMENT TO THE POST OF ASSISTANT DIRECTOR (TOWN PLANNING) UNDER URBAN DEVELOPMENT & POVERTY ALLEVIATION DEPARTMENT, GOVERNMENT OF MIZORAM.

(FEBRUARY-2026)

TECHNICAL SUBJECT PAPER-I

Time Allowed : 2 hours FM : 100

All questions carry equal mark of 1 each.
Attempt all questions.

1. Which theory explains urban growth in wedge-shaped sectors?

(a) Concentric Zone (b) Multiple Nuclei
(c) Central Place (d) Sector Theory

2. Lewis Mumford is best known for writing:

(a) The Death and Life of Great American Cities (b) Urban Land Economics
(c) The City in History (d) Tomorrow: A Peaceful Path to Real Reform

3. In the context of ancient Indian town planning, which of the following refers to a village layout shaped like a lotus?

(a) Dandaka (b) Sarvatobhadra
(c) Padmaka (d) Nandyavarta

4. Concentric Zone Theory was developed using studies of:

(a) Paris (b) London
(c) Chicago (d) Berlin

5. Patrick Geddes emphasized which planning sequence?

(a) Plan–Survey–Implement (b) Survey–Analysis–Plan
(c) Design–Build–Evaluate (d) Observe–Control–Expand

6. The concept of "Chowk" in traditional Indian towns primarily served as:

(a) Religious enclosure
(b) Military parade ground
(c) Public space for trade and social interaction
(d) Residential cluster

7. Tony Garnier proposed the:

(a) Garden City (b) Radiant City
(c) Linear City (d) Industrial City (Cité Industrielle)

8. Orthodox planning mainly stresses:

(a) Community control
(b) Market forces and social norms
(c) Expert-led rational decision making
(d) Political and traditional negotiation

9. Advocacy planning was proposed by:

(a) Pattrick Geddes
(b) Lewis Mumford
(c) Peter Hall
(d) Paul Davidoff

10. Systems approach views cities as:

(a) Static maps
(b) Interrelated subsystems
(c) Economic market hubs
(d) Political institutions

11. Mixed scanning model attempts to correct limitations of:

(a) Advocacy planning
(b) Pure rational planning
(c) Equity planning
(d) Structuralism

12. Ancient Mesopotamian cities were mainly organized around:

(a) Agricultural fields
(b) Markets
(c) Temples and palaces
(d) Military camps

13. Which of the following is considered a 'Primary Survey' in town planning?

(a) Collecting data from the Census of India
(b) House-to-house socio-economic interview
(c) Referring to existing satellite imagery
(d) Reviewing previous Master Plan documents

14. What does 'ECS' stand for in the context of transport planning and zoning regulations?

(a) Economic Car Space
(b) Equivalent Car Space
(c) Essential Commodity Space
(d) Environmental Clearance Standard

15. Which technique is used to estimate future population by assuming a constant percentage growth rate?

(a) Arithmetic Increase Method
(b) Geometric Increase Method
(c) Incremental Increase Method
(d) Graphical Method

16. 'Zoning Regulations' are primarily a tool for:

(a) Property tax regulation
(b) Improving city's economic growth
(c) Controlling use of land and density
(d) Limiting land development

17. The 'Origin-Destination (O-D) Survey' is a specific technique used in:

(a) Housing planning
(b) Transport planning
(c) Industrial planning
(d) Environmental planning

18. 'Non-conforming land use' refers to:

(a) Land that cannot be built upon
(b) Existing land use not in accordance with Master Plan zoning
(c) Government land use that need not follow Master Plan zoning
(d) Land that has no legal owner

19. According to URDPFI guidelines, for a town to be classified as a 'Hill Town,' it should generally be located at an altitude of more than:

(a) 300 meters
(b) 600 meters
(c) 800 meters
(d) 1,000 meters

20. Which index is commonly used to measure the 'Degree of Urbanization' in a region?

(a) Rate of perceived Urbanism
(b) Portion of urbanized area in percentage
(c) Percentage of total population living in urban areas
(d) Number of cars per household

21. Which Census organization in India is responsible for conducting the decennial census?

(a) NITI Aayog
(b) Survey of India (SoI)
(c) Office of the National Census Survey of India (ONCSI)
(d) Office of the Registrar General and Census Commissioner, India

22. In a GIS environment, 'Raster Data' is represented by:

(a) Lines and Points
(b) A grid of cells or pixels
(c) Text files
(d) Circular charts

23. Town Planning Scheme is mainly associated with:

(a) Brownfield development
(b) Greenfield development
(c) Central schemes in aid of implementation of Master Plan
(d) GIS-based Master Plan.

24. What is the primary purpose of a 'Screen Line Survey' in transport planning?

(a) To count passengers inside vehicles
(b) To detect double counting of trips and validate survey data
(c) To measure the speed of vehicles
(d) To check the license plates of cars

25. What is the standard height of a 'Mountain/High Curb' used in urban transport design?

(a) 5 cm
(b) 10 cm
(c) 20 cm
(d) 50 cm

26. What does 'PCU' stand for in traffic engineering?

(a) Private Car Usage
(b) Passenger Car Unit
(c) Public Carrier Utility
(d) Path Consistency Unit

27. A 'Dead-end' street that provides a turnaround area at the end is called a:

(a) Loop
(b) Alley
(c) Cul-de-sac
(d) Service Road

28. Which transport mode is generally the most energy-efficient for mass transit in high-density corridors?

(a) Private Cars
(b) Rail-based Transit (Metro/LRT)
(c) E-Buses
(d) Aggregators like Uber

29. In the 'Four-Step Model' of transport planning, which step involves the use of a 'Logit Model'?

(a) Trip Generation
(b) Trip Distribution
(c) Modal Split
(d) Traffic Assignment

30. As per URDPFI guidelines, the suggested Survey method for conducting intersection survey is:

(a) Cordon count
(b) Manual Count
(c) Video count
(d) Registration number counts survey

31. What is the 'Ruling Gradient' recommended by IRC for main roads in mountainous terrain like Mizoram?

(a) 2%
(b) 5%
(c) 10%
(d) 15%

32. Which of the following is a 'Demand-Side' travel management strategy?

(a) Widening an existing road
(b) Building a new flyover
(c) Congestion pricing
(d) Increasing the number of buses

33. In 'Origin-Destination' (O-D) surveys, the 'Desire Line' diagram represents:

(a) The direct path between origin and destination regardless of the existing road network
(b) The actual path taken by the vehicle from origin to destination
(c) The speed of vehicles on a specific road
(d) The line from origin to destination along the road taken.

34. A 'Cordon Survey' is typically conducted to:

(a) Measure the depth of the road pavement all the way to plot boundary
(b) Determine the volume and nature of traffic entering and leaving a defined area
(c) Count pedestrians in a park
(d) Determine the speed of traffic entering and leaving a defined area

35. 'Transit-Oriented Development' (TOD) focuses on:

(a) Moving all residents to transit oriented lands
(b) High-density, mixed-use development within walking distance of transit hubs
(c) Low-density, mixed-use development within walking distance of transit hubs
(d) Reducing the number of railway stations to improve speed

36. Which type of intersection control is most suitable for heavy traffic volumes on all approaching roads in a hilly urban area?

(a) Simple Stop signs
(b) Small Roundabouts
(c) Uncontrolled T-junctions
(d) Grade-separated interchanges

37. 'Paratransit' refers to:

(a) High-speed bullet trains
(b) Flexible transport services
(c) Public transport specially designed for PwDs
(d) Underground metro systems

38. According to URDPFI, the 'Walking Distance' to the nearest bus stop in an urban area should ideally not exceed:

(a) 200 meters
(b) 300 meters
(c) 4 kilometers
(d) 400 meters

39. Which survey is used to determine the 'Trip Rate' for different land uses?

(a) Speed and Delay survey
(b) Parking survey
(c) Road Inventory survey
(d) Home Interview Survey

40. In transport modeling, 'Zone Centroids' represent:

(a) The location of prime importance in a city where majority of travel commences
(b) The theoretical point where all trips in a traffic zone are assumed to start or end
(c) The geometric center of the city
(d) The densest part of the city to be pedestrianized

41. The 'Capacity' of a road is defined as:

(a) The maximum number of lanes it has
(b) The maximum hourly rate at which vehicles can reasonably be expected to traverse a section
(c) The total number of vehicles crossing a particular line of a road at a given time.
(d) The average number of vehicles traversing through the cordon point of the road within 15 minutes.

42. Which organization in India provides the standards for 'Geometric Design of Roads'?

(a) NITI Aayog
(b) Indian Roads Congress (IRC)
(c) Census of India
(d) Ministry of Road Transport & Highways (MoRTH)

43. The 'User Equilibrium' principle (Wardrop's First Principle) states that:

(a) No individual driver can reduce their travel time by unilaterally changing routes
(b) Total system travel time is minimized by choosing the road with the widest cross section
(c) All drivers must use the same lane at a given time
(d) Public transport is always faster than private cars

44. In hill road design, 'Extra Widening' of the carriageway at sharp curves is required due to:

(a) Thermal expansion of the road
(b) Off-tracking of rear wheels
(c) Accommodating street vendors
(d) Allowing for future road widening

45. In the 'Entropy Maximization Model' for trip distribution, the parameter β (beta) represents:

(a) The sensitivity of travelers to travel costs
(b) The total number of households
(c) The average traffic speed of the network
(d) The capacity of the destination zone

46. Under IRC norms for Hill Towns, the 'Maximum Gradient' allowed for a residential street in extreme cases is:

(a) 5%
(b) 7% to 8.5%
(c) 10% to 12.5%
(d) 25%

47. Which type of 'Traffic Assignment' accounts for the fact that as a road gets crowded, its travel time increases?

(a) All-or-Nothing Assignment
(b) Capacity Restrained Assignment
(c) Stochastic Assignment
(d) Incremental Assignment

48. A 'Unified Metropolitan Transport Authority' (UMTA) is recommended for cities to:

(a) Integrate the planning and management of various transport modes under one body
(b) Build and maintain metro rails to reduce private cars dependency
(c) Collect taxes for the central government
(d) Incorporate fast-track mechanisms for transportation in metro cities

49. 'Valley Curves' on hill roads must be designed primarily to satisfy:

(a) The comfort of the driver on corners
(b) Headlight sight distance during night driving
(c) Drainage of water
(d) Improve soil stability in corners

50. Machu Picchu of Inca Civilization is located in:

(a) Argentina
(b) Brazil
(c) Columbia
(d) Peru

51. The concept of 'Conurbation' was given by:

(a) Lewis Mumford
(b) Homer Hoyt
(c) Patrick Geddes
(d) C.D. Harris

52. The concept of Urbanism as a way of life was developed by:

(a) Thomas Robert Malthus
(b) Karl Marx
(c) Louis Wirth
(d) Lewis Coser

53. Name the theory developed by Chauncy Harris and Edward Ullman in analysing a city.

(a) Ecological theory
(b) Zone theory
(c) Sector theory
(d) Multiple nuclei theory

54. Participatory planning is an urban planning paradigm that involves:

(a) Politicians
(b) People
(c) Town Planner
(d) Prominent citizens

55. The cities of Mohenjo-Daro and Harappa came into being in the valley of Indus river around _____.

(a) 750 BC
(b) 1000 BC
(c) 2500 BC
(d) 1500BC

56. Who is regarded as the first town planner and 'inventor' of the orthogonal urban layout?

(a) Aristotle (b) Hippodamus

(c) Copernicus (d) Kent

57. The building activity expands in a natural way along the sides of main road, this type of development can be called as:

(a) Ribbon development (b) Concentric spread

(c) Satellite Growth (d) Scattered growth

58. In how many types were the villages classified in Manasara?

(a) 7 types (b) 8 types

(c) 6 types (d) 4 types

59. Ecumenopolis means:

(a) City of the dead (b) A fortified city

(c) A capital city of a country (d) A city made of whole world

60. The term coined in 'Brutland Report' is:

(a) Resilience (b) Ecology

(c) Sustainability (d) Environment

61. Geddian Triangle is based on:

(a) Environment approach (b) Social approach

(c) Capitalist approach (d) Political approach

62. Which of the following books Ebenezer Howard published in 1898 ?

(a) Tomorrow : A Peaceful Path to Real Reform

(b) Garden Cities of Tomorrow

(c) Model Garden Cities of Tomorrow

(d) Garden Suburb of London.

63. The application platform developed by Indian Space Research Organization (ISRO) allowing representation of Earth's surface like Google Earth is called:

(a) IRNSS (b) Bhuvan

(c) Bing Map (d) Open Street Map

64. As per URDPFI guidelines, the developed area average densities for large cities in hill areas is :

(a) 75-125 pph (b) 45-75 pph

(c) 60-90 pph (d) 125-175 pph

65. Delphi method is a:

(a) Perception of respondents

(b) Study technique of natural resources

(c) Forecasting method based on experts' opinion

(d) Technique of assessment of ecosystem resources

66. In GIS, DEM represents information on:

(a) Geology
(b) Topography
(c) Land-cover
(d) Hydrology

67. SDG 11 pertains to:

(a) Affordable and Clean Energy
(b) Climate Action
(c) Clean water and Sanitation
(d) Sustainable Cities and Communities

68. _____ techniques are used for anticipating future which is a necessary step in the planning process.

(a) Formulation
(b) Correlation
(c) Rejection
(d) Projection

69. Durkheim's concept of organic solidarity is found in a:

(a) Agricultural society
(b) Tribal society
(c) Modern society
(d) Traditional society

70. As per the URDPFI Guidelines, choose the option which indicate the appropriate hierarchy of plans from higher to lower order:

(a) Perspective Plan > Development Plan > Local Area Plan
(b) Development Plan > Special Purpose Plan > Annual Plan
(c) Local Area Plan > Annual Plan > Development Plan
(d) Special Purpose Plan > Perspective Plan > Local Area Plan

71. Which of the following can be determined through Remote Sensing?

1. Population Density
2. Tree cover
3. Agricultural land
4. Lakes and wetlands
5. Transportation network

Which of the above statements are correct?

(a) 1, 2, 4 and 5
(b) 2, 3, 4 and 5
(c) 1, 3, 4 and 5
(d) 1, 4 and 5

72. Under the current urban development programmes/missions, GIS based master plan is a component under which of the following?

(a) Smart Cities Mission
(b) AMRUT Mission
(c) HRIDAY
(d) None of the above

73. The boundaries of individual land parcels of land (lots) is precisely defined in:

(a) Zonal Plan
(b) Master Plan
(c) Comprehensive development plan
(d) Perspective plan

74. As per NUIS- National remote Sensing Centre , which of the following is most appropriate match:

No.	I	No.	II
A	Regional Plan	i	1:10,000
B	Development Plan	ii	1:50,000
C	Sub-Regional Plan	iii	1:2000
D	Zonal Plan	iv	1:25,000
		v	1:1,00,000

(a) A-v,B-ii,C-iii,D-iv
(b) A-ii,B-i,C-iv,D-iii
(c) A-ii,B-iv,C-i,D-iii
(d) A-v,B-i,C-iv,D-iii

75. The type of surveying in which the shape of the earth is taken into account is _____.

(a) Topographic Surveying
(b) Hydrographic Surveying
(c) Geodetic Surveying
(d) Plane Surveying

76. Zoning in urban and regional planning refers to:

(a) Creating a single type of land use.
(b) Designating areas for specific functions like residential, commercial, etc.
(c) Developing only green spaces.
(d) Ignoring land use regulations

77. Name the traffic survey data which is plotted by means of 'desired lines.'

(a) Accident
(b) Origin and destination
(c) Classified volume
(d) Speed and delay

78. A 'cul-de-sac' is a street where:

(a) Only two-wheelers are permitted
(b) Through traffic is discouraged
(c) Pedestrians are not permitted
(d) Vehicles are permitted to move in one direction only

79. A Street, which serves or is designed to serve as a traffic way for a neighbourhood or as a feeder to a major roadway:

(a) Sub arterial road
(b) Arterial road
(c) Collector Road
(d) Local Road

80. Sequentially arrange the following as per four stage model for transportation planning

1. Trip generation
2. Mode choice
3. Trip distribution
4. Route choice

(a) 1,2,3,4
(b) 1,3,2,4
(c) 1,4,2,3
(d) 1,3,4,2

81. V7 concept given by Le Corbusier refers to:

(a) Neighbourhood Planning
(b) Housing Typologies
(c) Architecture Design Principle
(d) Hierarchy of Roads

82. Which term describes the redevelopment of old, run-down city areas?

(a) Suburban sprawl
(b) Gentrification
(c) Urban renewal
(d) Ruralization

83. Scientific planning of transportation system and mass transit facilities in cities should be based on:

(a) Spot speed data
(b) Origin and destination data
(c) Traffic volume data
(d) Accident data

84. Transit Oriented Development:

(i) reduces urban sprawl
(ii) will induce shift from non-motorized to motorized modes of travel
(iii) highly depends on establishing mixed land use zone as a part of strategic densification

Which of the above statements are true?

(a) Only (i)
(b) (i) and (iii)
(c) (i) and (ii)
(d) (i), (ii) and (iii)

85. The maximum number of points of conflicts in two way two lane roads meeting at right angles is:

(a) 12
(b) 32
(c) 4
(d) 24

86. Camber (cross slope) is primarily provided in roads for?

(a) Water drainage
(b) Road marking
(c) Speed control
(d) Additional lane

87. Namo Bharat trains is a type of:

(a) Light rail transit
(b) Sub urban train
(c) Rapid rail transit
(d) Bus road transit system

88. In urban transport sector, SLB means:

(a) Service Level Benchmarks
(b) System Level Benchmarks
(c) Station Level Benchmarks
(d) Standard Level Benchmarks

89. The mobility of people and goods can be maximized by :

(a) Minimizing travel time
(b) Provide adequate frequency of service, capacity, safety & reliability
(c) Minimizing travel cost
(d) All of the above

90. _____ indicate the boundary between the carriage way and the shoulder or islands or footpaths.

(a) Kerbs
(b) Sideway
(c) Margin
(d) Gutter

91. What is the full form of ITS?

(a) Intellectual Transit Structure
(b) Intelligent Transport Structure
(c) Intellectual Transport System
(d) Intelligent Transportation System

92. Select the correct statement(s) from the following:

1. Introduction of automobiles led to urban sprawl
2. Compact cities show relatively higher carbon emissions
3. Land use and transportation planning is inter-dependent on each other
4. Addition of a transport mode in an urban area does not change accessibility

(a) 1 and 2
(b) 2 and 3
(c) 1 and 3
(d) 3 and 4

93. Select the correct sequence of steps for designing the operation of a signalized intersection.

(a) Signal Phasing ⟶ Green Allocation ⟶ Cycle Length Selection
(b) Green Allocation ⟶ Cycle Length Selection ⟶ Signal Phasing
(c) Cycle Length Selection ⟶ Signal Phasing ⟶ Green Allocation
(d) Signal Phasing ⟶ Cycle Length Selection ⟶ Green Allocation

94. Speed and delay studies along routes are best done using the following method

(a) Moving Observer Method
(b) Registration Number Method
(c) Elevated Observer Method
(d) Photographic method

95. The median separating dual carriageways is also referred to as:

(a) Shoulder
(b) Mid reservation
(c) Central reservation
(d) Split reservation

96. In the context of Rural Road Development Plan; PMGSY refers to:

(a) Pradhan Manti Gram Sadak Yojna
(b) Pracheen Mukya Gram Sadak Yojna
(c) Pradhan Mantri Gram Seva Yojna
(d) Pradhan Mantri Gram Sangh Yojna

97. In a traffic survey, Enoscope is used to measure:

(a) Volume to Capacity ratio
(b) Sight distance
(c) Spot speed
(d) Intersection delay

98. Which of the following initiatives of the Government of India is also known as the National Master Plan for Multi-modal Connectivity?

(a) PM Gati Shakti
(b) Bharatmala
(c) Parvatmala
(d) Sagarmala

99. IRC recommends a minimum desirable width of _____ for medians of rural highways.

(a) 3.0 m
(b) 5.0 m
(c) 3.5 m
(d) 4.5 m

100. The imaginary lines dividing the study area taken for survey into parts is called as:

(a) External cordon lines
(b) Internal cordon lines
(c) Desire lines
(d) Grid lines

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